

**San Jose State University  
Department of Electrical Engineering**

**Mini-Project III – One-Level Cache Simulator (Due Apr. 30)**

In this project, you are to design a data cache simulator. For the design language, you may use any of your choice; C/C++, C#, Java, Matlab, Python, Verilog, or VHDL. The simulator you implement needs to work for  $m$ -way set-associative cache, where  $m = 1, 2, 4, 8$ , or fully associative. The size of the cache should be 16KB, 32KB, 64KB, or 128KB. LRU (Least Recently Used) or FIFO or Random is to be the cache replacement algorithm for selecting the block to be replaced in the set. All the memory in this project will be byte addressable. The block size should be 16, 32, 64, or 128 bytes per block.

Your simulator needs to support the following 3 cache design parameters: cache size, set associativity, block/line size, and different cache replacement algorithm.

The “address trace” is provided in the separate file, “addr\_trace.” The file contains the trace of load instructions executed for some program. Each line represents a byte address referenced by an executed load instruction. To help compress the traces, instead of storing the absolute memory address, we store the difference or offset from the last address in the trace (aka differential coding). You should have the start of the trace you see as follows, and then use that trace for your cache simulator.

Here is the start of the trace file "addr\_trace" you will be working with:

```
536869800
16
536970168
27984
331676
-29388

-537300560
536970136
389812
-33000
```

Once restored (make sure you see the same load addresses) is:

```
536869800
536869816
1073839984
1073867968
1074199644
1074170256
536869696
1073839832
1074229644
1074196644
```

Once your simulator completes execution for the given input for the cache design parameters as described before, the output should include: total number of simulation addresses, total number of cache hits, and cache hit ratio.

The maximum values your simulator needs to support for each of the cache design parameters is 128KB for the cache size, 128 bytes for the block size, and the full associativity.

### **What to Submit**

For this project, GUI is not necessary. You will upload to Canvas the source code file of your simulator and a project report file with three required graphs I recommend that you consider to have the direct mapped cache working first. Three graphs, say by Excel or Matlab, are required together with the data analysis and evaluation in your project report file:

For the first one, a line will be drawn; the Y-axis for the graph will be the miss rate, and the X-axis for the graphs will be the size of the cache (given the fixed block size 32; fixed associativity 4). For results, you will need to run your cache simulator for cache sizes of 16 KB, 32KB, 64KB, and 128KB of data.

For the second one, a line will be drawn; the Y-axis for the graph will be the miss rate, and the X-axis for the graphs will be the size of the block (given the fixed cache size 8192; fixed associativity 4). For results, you will need to run your cache simulator for block sizes of 16, 32, 64, and 128.

For the third one, a line will be drawn; the Y-axis for the graph will be the miss rate, and the X-axis for the graphs will be the associativity (given the fixed cache size 8192; fixed block size 32). For results, you will need to run your cache simulator for associativity of 1, 2, 4, 8, and the fully associative.

For the project report writing guideline, please see the guideline posted on the Canvas site.

Also, please note that a 1-page progress report is to be submitted to Canvas on Apr 19.